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AT THE ABB Smart Power factory in Nelamangala, Bengaluru, the quality inspection process is no longer manual. An artificial intelligence (AI)-enabled camera takes pictures of the finished moulded case circuit breaker (MCCB) from five sides and checks against the algorithm for defects such as terminal thickness, height, printing errors and scratches. Only when all the parameters are met—no smudged text, printing error or structural damage—is the MCCB cleared for packaging. Saju SR, Senior Vice President, Smart Power Division, ABB India, claims zero returns since the system was implemented. The system has brought a level of precision that is impossible for a human being.

ABB is not alone. AI—from being focused largely on software functions—is finding its way to physical environments such as factory shop floors and warehouses as well. At the Tata Steel factory in Jamshedpur, an AI-enabled sensor tests for corrosion in pipes, a change from the earlier system where the operator had to decide whether to continue running the system based on the camera feed. Now, the AI model determines the extent of the corrosion and takes a call if it's safe to run the machine. The 118-year-old company runs 650 AI models at manufacturing locations; over 500 are in India. It has also set up a remote maintenance centre to monitor mining operations. "Our Integrated Remote Operations Centres, Raw Material Supervision Centres and Integrated Maintenance Excellence Centre provide intelligent, location-agnostic oversight of processes from hundreds of kilometres away. This enables centralised predictive and prescriptive maintenance, saving over 1,500 hours in asset maintenance

time," says Jayanta Banerjee, Chief Information Officer, Tata Steel.

To be sure, smart automation has been transforming Indian shop floors for over a decade, so much so that robot-powered assembly lines and warehouses are a common sight across the country. However, physical AI goes beyond smart automation. The difference is the intelligence layer. "In smart automation, there is no decision-making by the machine. It's a pre-programmed algorithm operating on the basis of inputs provided in the past," says Saju. "With physical AI, you add an AI layer to physical systems so that they can have the cognitive abilities of a human being." This gives the system the ability to understand what is happening and take a decision based on the input received. The ability to process device data with AI locally on the physical device is an essential component of physical AI, also sometimes referred to as Edge AI.

L&T Technology Solutions, which works with companies to set up digital twins, or virtual replicas of their factories, has developed an AI perception system that allows vehicles to make split-second navigation and safety decisions without relying on cloud connectivity. It has created plant-level twins being used by global OEMs to identify micro-efficiencies in assembly lines, says Amit Chadha, Managing Director & Chief Operating Officer, L&T Technology Services. "This has led to a demonstrated 10-12% productivity gain. This evolution, from descriptive to prescriptive and predictive twins, has been powered by AI," he says.

AUTOMATION INITIATIVES HAVE TRADITIONALLY FOCUSED ON JOBS THAT ARE REPETITIVE OR UNSAFE

Earlier this year, a report by the World Economic Forum and the Boston Consulting Group, *Physical AI: Powering the New Age of Industrial Automation*, said "recent innovations in software and hardware have ushered in a step-change in robotic capability, enabling robots to perform complex tasks in dynamic environments with simpler deployment. Advances in AI and complex simulations, enabled by accelerated computing using graphics processing units (GPUs), have made it feasible to run AI models and algorithms in real time, unlocking new applications. This AI-based approach focuses on enabling robots to perceive, plan and act in complex, real-world scenarios, effectively achieving a level of physical intelligence."

Automation initiatives have traditionally focused